One of the most powerful and far-reaching theoretical models ever devised, already introduced in these pages, is Einstein's general theory of relativity—but you can call it GR after you get to know it better. Published in 1916, GR outlines the relevant mathematical details of how everything in the universe moves under the influence of gravity. Every few years, lab scientists devise ever more precise experiments to test the theory, only to further extend the envelope of the theory's accuracy. A modern example of this stunning knowledge of nature that Einstein has gifted us, comes from 2016, when gravitational waves were discovered by a specially designed observatory tuned for just this purpose.[†] These waves, predicted by Einstein, are ripples moving at the speed of light across the fabric of space-time, and are generated by severe gravitational disturbances, such as the collision of two black holes.

And that's exactly what was observed. The gravitational waves of the first detection were generated by a collision of black holes in a galaxy 1.3 billion light-years away, and at a time when Earth was teeming with simple, single-celled organisms. While the ripple moved through space in all directions, Earth would, after another 800 million years, evolve complex life, including flowers and dinosaurs and flying creatures, as well as a branch of vertebrates called mammals. Among the mammals, a sub-branch would evolve frontal lobes and complex thought to accompany them. We call them primates. A single branch of these primates would develop a genetic mutation that allowed speech, and that branch—*Homo sapiens*—would invent agriculture and civilization and philosophy and art and science. All in the last ten thousand years. Ultimately, one of its twentieth-century scientists would invent relativity out of his head, and predict the existence of gravitational waves. A century later, technology capable of seeing these waves would finally catch up with the prediction, just days before that gravity wave, which had been traveling for 1.3 billion years, washed over Earth and was detected.

Yes, Einstein was a badass.

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When first proposed, most scientific models are only half-baked, leaving wiggle room to adjust parameters for a better fit to the known universe. In the Sun-based "heliocentric" universe, conceived by the sixteenth-century mathematician Nicolaus Copernicus, planets orbited in perfect circles. The orbit-the-Sun part was correct, and a major advance